4378310: KD Bus Sites Production VPN Migration

CR Number	CR-4378310	Request Date	2026-02-02
Requested By	Olli Ojala	Risk Level	MEDIUM
Environment	Meraki Dashboard - KD BUS | Azure East & South East	Rollback	< 5 min/site
Total Sites	24 sites (71 subnets)	Downtime	< 5 min/site during route switch

1. Summary

Purpose: Migrate 24 KD Bus sites from SonicWall VPN to Azure-hosted FortiGate firewalls via IPsec VPN. Migration follows successful pilot test (CR-4342297).

Strategy: Phased rollout - Pilot site first, then 2 production batches.

2. Migration Phases

Phase	Sites	Subnets	Description
Phase 1 - Pilot	1	2	Malaga-Morley (validation)
Phase 2 - Batch 1	11	43	NSW, QLD, selected SA sites
Phase 3 - Batch 2	12	26	Remaining SA, all WA sites

3. Infrastructure

Component	Azure East (Primary)	Azure South East (DR)
FortiGate Public IP	68.218.113.249	4.200.139.87
FortiGate Inside IP	10.200.1.68	10.201.1.68
Route Table	KD-Prod-RouteTable	KD-Mel-RouteTable
Associated Subnets	East workload subnets (10.200.x.x)	SEast workload subnets (10.201.x.x)

4. Design / Traffic Flow

Primary/DR Architecture: Azure East is the **primary region** for production traffic. Azure South East is the **DR region** (pre-staged, not active under normal operation).

Route Table Isolation: KD-Prod-RouteTable is associated only with East workload subnets; KD-Mel-RouteTable only with SEast workload subnets. Each region routes to sites via its local FortiGate. This prevents asymmetric routing during normal operation.

Why routes in both regions: Site subnets are pre-staged in both route tables so that, during a DR event, workloads failed over to SEast retain connectivity to all site subnets via the SEast FortiGate.

DR Failover: Manual failover (not automatic). DR is invoked by Azure workload failover (ASR/manual), not by this VPN configuration. *DR failover testing is not in scope for this CR; DR readiness is achieved by pre-staging configuration only.*

5. Phase 1 - Pilot Site Details

Site	Network 1	Network 2
Malaga-Morley	`10.9.52.0/24`	`10.91.52.0/24`

6. Changes to be Made

System	Action
FortiGate East	Configure IPsec Phase1/Phase2 for each site, create address objects, firewall policies
FortiGate SEast	Configure IPsec Phase1/Phase2 for each site, create address objects, firewall policies
Meraki	Remove network tag restriction from IPsec peers (staged per batch, not all sites at once)
Azure East	Add routes to KD-Prod-RouteTable **per site** as migrated (next-hop: 10.200.1.68)
Azure SEast	Add routes to KD-Mel-RouteTable **per site** as migrated (next-hop: 10.201.1.68) - DR pre-staging

Route Staging: Routes are added per site during migration, not all 71 subnets at once. Total routes at completion: 71 per route table (142 total).

7. IPsec VPN Parameters

Parameter	Value
IKE Version	IKEv2
Encryption	AES-256
Authentication	SHA256
DH Group	14 (2048-bit)
PFS (Perfect Forward Secrecy)	Enabled, Group 14
Phase 1/2 Lifetime	1800s / 3600s
DPD (Dead Peer Detection)	Enabled (on-demand)
NAT-T	Enabled (auto-detect)
PSK	Shared securely outside CR (>32 chars)

Firewall Policies: FortiGate firewall policies permitting tunnel traffic to/from Azure backend are confirmed in place (prerequisite).

8. Site Subnet Summary

Site	Subnets	Phase
Malaga-Morley	`10.9.52.0, 10.91.52.0`	Pilot
Belmont North	`10.21.52.0, 10.2.52.0, 10.24.52.0`	Batch 1
Brookvale	`10.2.64.0, 10.21.64.0, 10.28.64.0, 10.25.64.0, 10.26.64.0, 10.30.64.0, 10.31.64.0`	Batch 1
North Sydney	`10.2.68.0, 10.21.68.0, 10.28.68.0, 10.25.68.0, 10.26.68.0, 10.30.68.0, 10.31.68.0`	Batch 1

Mona Vale	`10.2.66.0, 10.21.66.0, 10.28.66.0, 10.25.66.0, 10.26.66.0, 10.30.66.0, 10.31.66.0`	Batch 1
Wickham	`10.2.60.0, 10.26.60.0, 10.21.60.0`	Batch 1
Queens Wharf	`10.21.54.0, 10.2.54.0`	Batch 1
Clontarf	`10.71.52.0, 10.7.52.0, 10.75.52.0`	Batch 1
North Lakes	`10.71.56.0, 10.7.56.0`	Batch 1
Hamilton	`10.21.50.0, 10.24.50.0, 10.2.50.0, 10.27.50.0, 172.20.120.0`	Batch 1
Goolwa	`10.81.72.0, 10.8.72.0`	Batch 1
Murray Bridge	`10.81.58.0, 10.8.58.0`	Batch 1
Mt Barker	`10.81.62.0, 10.8.62.0`	Batch 2
Aldgate	`10.81.64.0, 10.8.64.0`	Batch 2
Pooraka	`10.81.54.0, 10.8.54.0`	Batch 2
Willaston	`10.81.68.0, 10.8.68.0`	Batch 2
Nuriootpa	`10.81.60.0, 10.8.60.0`	Batch 2
Welshpool	`10.91.64.0, 10.9.64.0, 10.95.64.0`	Batch 2
Walliston-Kalamunda	`10.91.66.0, 10.9.66.0, 10.9.67.0`	Batch 2
Bayswater	`10.91.58.0, 10.9.58.0`	Batch 2
Forrestfield	`10.41.1.0, 10.91.41.0`	Batch 2
Redcliffe	`10.9.68.0, 10.91.68.0`	Batch 2
Geraldton	`10.91.70.0, 10.9.70.0`	Batch 2
Henley Brook	`10.91.54.0, 10.9.54.0`	Batch 2

9. Test Plan (Per Site)

Normal State Acceptance:

- **Site → Azure East (Primary):** Ping from site to 10.200.10.10 - expect reply
- **Azure East → Site:** Ping from Azure East VM to site subnet - expect reply
- RDP to Azure East VMs to confirm bidirectional traffic (no asymmetric routing)
- Verify East FortiGate tunnel stats incrementing (production traffic)

DR Pre-staging Validation:

- **Site → Azure SEast (DR):** Ping from site to 10.201.10.61 - expect reply (tunnel up)
- Verify SEast FortiGate tunnel established (idle/standby - no production traffic expected)
- *Note: DR failover testing (forced East tunnel down) is not in scope for this CR*

Isolation Check:

- Confirm other sites unaffected (Meraki tag changes are staged per batch)

10. Rollback Procedure (Per Site)

If tests fail: Execute in order (estimated < 5 minutes per site)

1. Remove site routes from KD-Prod-RouteTable (Azure East)
2. Remove site routes from KD-Mel-RouteTable (Azure SEast)
3. Disable VPN tunnels on both FortiGates
4. Verify site connectivity via existing SonicWall

Why both regions: Rollback removes both primary and DR paths to force reversion to SonicWall and prevent partial failover/asymmetric routing.
SonicWall Fallback: SonicWall configuration remains unchanged until completion of all batches + 14 days soak period.

11. Prerequisites

- CR-4342297 (Pilot Test) completed successfully
- FortiGate address objects created for all 71 subnets
- FortiGate firewall policies in place (tunnel ↔ Azure backend)
- NSG permits FortiGate inside IPs (10.200.1.68, 10.201.1.68)
- Meraki Dashboard access with Org write permissions
- PSK shared securely (>32 characters)

Full Runbook: CR-2026-002-KD-Bus-Production-VPN-Migration.html
Dependency: CR-4342297 (IPsec Pilot Test) must be completed first
DR Scope: DR failover testing is not in scope; DR readiness achieved by pre-staging configuration only